



Active Volcano on Mars

Tucson astronomers have discovered semi-parallel fissures on Mars, indicating recent volcanic activity. <https://digit.in/dec20-27>



Arecibo Observatory demolition

Following breakage of two support cables, the Puerto Rican observatory is set to be dismantled. <https://digit.in/dec20-28>

PROTOCOLS FOR ALIEN CONTACT

What are the official rules for interacting with extraterrestrials?

Aditya Madanapalle | aditya@digit.in

T

he first rule of interacting with the aliens, is don't. The Outer Space treaty is the basis of the

international space law, and is signed by 104 nations, including all the spacefaring nations. The treaty has widespread international support, and its provisions are binding on all states, so it is considered to be at par with international law. Article IX of the treaty deals specifically with alien contact, but not in the sense of how we are supposed to go about saying our hellos. Instead, its focus is to ensure that in the search for life, we do not end up introducing it to planets such as Mars, and compromising our own scientific investigation. Article IX of the treaty says,

"States Parties to the Treaty shall pursue studies of outer space, including the Moon and other celestial bodies, and conduct exploration of them so as to avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter and, where necessary, shall adopt appropriate measures for this purpose."

AVOIDING CONTAMINATION

Article IX serves as the legal basis for the meeting of the Committee on Space Research (COSPAR) every two years, which has representatives from the scientific community that come up with recommendations to ensure that the objectives of the treaty are met.

Scientists in the early days of the space age began worrying that careless exploratory probes from Earth with microbes going to the Moon or Mars might contaminate the planets, and compromise future investigations dedicated to finding life on these worlds. To prevent such contamination, the Interna-

tional Council of Scientific Unions established the Committee on Contamination by Extraterrestrial Exploration (CETEX), which then transformed into the Committee on Space Research (COSPAR). COSPAR regularly updates its recommendations for planetary protection and planetary quarantine. The guidelines depend on the type of mission, for example probes meant to observe the Sun or Mercury do not need to be very careful, as any microbes on these spacecraft are not likely to contaminate anything. The missions are divided into categories from Category I to category V, with an increasing amount of exposure to complications following contamination. Only missions to Mars, Europa and Enceladus are considered to be in category III and IV, although Titan which falls in category II missions has the necessary ingredients to support not only life as we know it on Earth, but also another form based on hypothetical biochemistry. Category V is divided into two parts, unrestricted and restricted. The unrestricted missions are